Orthogonality, antipodes, and a depth-32 falsification map for white-box mean estimation

Challenge: ARC White-Box Estimation Challenge 2026, Phase 1

Referenced graded submission: #325459 (`pure-0.2.1s`; adjusted **3.202e-07**, secondary/raw **8.075e-07**)

Also nominate (same estimator): #325112

Code: branch `exp/l31-specialist`, estimator `estimator.py` (`VERSION = "pure-0.2.1s"`)

Date: 2026-08-07

LLM disclosure. Research direction, experiment design, coding, and drafting of this write-up were AI-assisted (Cursor / Composer). All quantitative claims below are tied to committed probe scripts under `experiments/probes/` or official `whest run` / AIcrowd graded submissions. Where a claim is interpretive rather than measured, it is marked as such.

Abstract

Under the Phase-1 budget $B = 2.72\times10^{11}$ (depth 32, width 256, ReLU MLP), we find that **final-layer post-ReLU mean error is almost entirely Monte Carlo variance**, not correctable bias. A structured spherical sampler — **Sylvester–Hadamard orthonormal frames**, random row permutation, **column signs**, **antipodal pairing**, and **χ-mean radius** — reaches online adjusted score **3.202e-07** at utilization ≈0.40. A large falsification map (column-sign redesigns, partial ONBs, robust aggregators, James–Stein shrinks, last-layer Gaussian closures, residual MLPs, mid-layer recenter, and post–v0.10 residual packs) fails to beat this baseline once scored under the official adjusted metric and seed protocol 3.0. Independently, a trajectory-calibrated Hermite moment chain (port of public #314695) improves plain kprop but remains **~6× worse in raw MSE** than Hadamard sampling on Mini. We argue the Phase-1 sampling ceiling is architectural: cheap white-box correctors do not see a usable residual at L31, so further gains require better *deterministic* propagation (Phase-2 axis), not more sampler cosmetics.

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1. Problem and scoring

We estimate $\mu_{31} = E[\mathrm{ReLU}(z_{31})]$ for each of 256 units of a depth-32 width-256 bias-free ReLU MLP, given white-box weights and FLOP budget B. The ranked metric is

> adj = mean_m(MSE_m · max(0.1, C_m / B)),

> C_m = F_m + λ R_m, $\lambda = 10^{11}$.

All offline Mini gates use `seed_protocol` **3.0** (`MLP.from_row(..., seed_protocol_version="3.0")`). Early probes that fed raw `mlp_seed` into estimators inflated offline MSE by ~1.24× and are discarded.

Environment note (2026-08): evaluators run **flopscope 0.10 / whestbench 0.14** (dtype-aware costs, billed data movement, participant code limited to one physical core). Our Mini-100 under v0.10: adj **2.928e-07**, util ≈0.397; online public/private suite for #325459 remains **3.202e-07**.

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2. Method: Hadamard frames + radius (`pure-0.2.1s`)

2.1 Construction

Let $H \in \{\pm 1\}^{d \times d}$ be the Sylvester Hadamard matrix ($d = 256$). Rows of $H/\sqrt{d}$ form an ONB. For each frame $t = 1, \ldots, T$:

1. Draw a uniform row permutation π_t and column signs $s_t \in \{\pm 1\}^d$.
2. Form $M_t = (H_{\pi_t} / \sqrt{d}) \odot s_t$ (row perm + column signs; still a signed ONB).
3. Scale by the χ-mean radius $e_r = \mathrm{E}[\| g \|_2]$ for $g \sim \mathrm{N}(0, I_d)$.
4. Emit the antipodal pair $\{+e_r M_t, -e_r M_t\}$.

Forward all $N = 2Td$ directions through the MLP in float32; return the empirical mean of the final-layer ReLU activations (zeros for earlier layers). Sample count targets utilization ≈0.40 via an analytical FLOP model (matmul $2d^2$ per layer per sample + ReLU).

2.2 Why this structure (mechanistic content)

Property	Role
ONB rows	Exact second angular moment of the direction set (uniform energy on coordinates).
Antipodes	Exact cancellation of odd moments of the *input* direction measure; removes a large class of first-order biases before the first ReLU.
χ radius	Matches $\mathrm{E}[\| x \|_2]$ of isotropic Gaussians without sampling radii.
Random perm + signs	Breaks alignment with any fixed weight eigenframe while preserving ONB geometry.

Interpretive claim (validated indirectly): under full ONB packing, further "design" of signs/perms cannot change the *set* of rows up to signed permutation of an ONB — only the *randomization* of which ONB is used. Empirically, structured sign designs never beat i.i.d. signs (Section 3.1).

2.3 Scores

Artifact	Mini adj	Online adj
`pure-0.2.1s`	2.928e-07 (v0.10 Mini-100)	**3.202e-07** (#325459 / #325112 / #324164)
Prior gain-blend λ=0.97	2.889e-07	3.302e-07 (#324226) — Mini↑ online↓
Prior util=0.39	2.929e-07	3.301e-07 (#324236) — Mini↑ online↓

We therefore **lock util≈0.40** and refuse Mini-only util/gain knobs.

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3. Falsification map (depth 32)

All comparisons below are against the same Hadamard baseline under seed protocol 3.0. Ratios are baseline_mse / method_mse (higher is better for the method; <1 means method loses).

3.1 Geometry of the direction set

Intervention	Evidence	Verdict
Column-sign designs (balanced / Hadamard-colperm / random)	Mini-100: baseline raw **7.383e-07**; best alt (hada_colperm) **7.845e-07** (ratio 0.94) — `hada_sign_focused_final.json`	**Kill**
Within-frame sign balance / block4 / pair_anti	Mini-21: all worse than i.i.d. — `hada_sign_balance_final.json`	**Kill**
Partial ONB (fewer rows → more frames)	Mini-21: r=256 best; r=64 ratio 0.56 — `hada_partial_onb_final.json`	**Kill**; lock full ONB

3.2 Aggregation and shrinkage

Intervention	Evidence	Verdict
Trim / winsor / median	Mini-100: mean best; win05 ratio 0.06 — `hada_agg_robust_final.json`	**Kill** (ReLU skew → mean required)
James–Stein / soft-gain	Mini-100: ≈ noise (ratio ≈1.000) — `hada_js_shrink_final.json`	**Kill**

3.3 Analytic closures and learned residuals

Intervention	Evidence	Verdict
Last-layer diag-Gaussian / blends	Mini-31: blend05 ratio 0.44 — `hada_last_gauss_final.json`	**Kill**
L30/L31 residual MLP / ridge on $(Y_{31} - \square\mu)$	Full×250 OOS $R^2 \approx 0.001$; features ≈ noise vs Hadamard residual	**Kill**
Hard L30 recenter from noisy pilot	Extremely brittle: ≥5% relative RMS noise destroys ~50× oracle gain	**Kill** for online
Trajectory Hermite chain (baked #314695 coefs)	Mini-30: chain raw **7.15e-06** vs Hadamard **1.14e-06** (hada/chain ≈0.16) — `moment_chain/artifacts/eval_mini_gate.json`	Better than plain kprop, **not** competitive with sampling
Atlas DAgger + two-pass residual/abs-refit	Pass-2 **numerically identical** to pass-1 on Mini+MC and on atlas true M_{11} (24/6) — `dagger_atlas_twopass_meta.json`, `dagger_mini_twopass_meta.json`	**Kill** two-pass (redundant; matches public #314695 README)

True-cumulant ceiling (atlas connected pre κ_3/κ_4/k_{iij}, baked coefs)	Val **2.92e-06** vs MC-probe **4.36e-06** (~1.5×); still ≫ Hadamard — `true_cumulant_ceiling.json`	Probe noise ~1.5×; **family floor ~3e-06** even with oracle cumulants

3.4 Bias–variance (corrected; flopscope 0.10)

On Mini-20 (seed protocol 3.0), two independent full-*N* `pure-0.2.1s` streams $Y^{(1)}$, $Y^{(2)}$ with averaged predictor $\bar{Y} = \frac{1}{2}(Y^{(1)} + Y^{(2)})$:

Quantity	Value
mean MSE_single	6.46×10^{-7}
mean MSE_$\bar{Y}$	3.07×10^{-7}
MSE_$\bar{Y}$ / MSE_single	**0.475** (≈½)
bias² ≈ 2·MSE_$\bar{Y}$ − MSE_single	-3.2×10^{-8} (slightly negative ⇒ consistent with ≈**0**)

Artifact: `writeups/artifacts/bias_var_mini20_v010.json`.

Interpretive conclusion: at this budget the estimator is nearly unbiased; the adjusted score is a variance–utilization tradeoff. This explains why bias-facing machinery (θ correctors, JS, recenter) fails, and why Mini-tuned util/gain often regresses online.

3.5 Residual / off-meter paths (post one-core lock)

After flopscope v0.10's one-physical-core residual safeguard (forum #18125), we re-tested RemoteArray detach + shipped OpenBLAS packs (`experiments/pure021s_rp_ob/`). Working detach is `tolist()` (RPC); soft probes (`pickle`, `memoryview`, `dlpack`, `tofile`, …) fail smoke. Online results:

Path	Online adj	Verdict
Best stable residual pack (ob9–ob12)	~3.47–3.53e-07	**Worse** than honest 3.202e-07
Optimistic packing / rate-gates (ob13–19)	0.23–0.68 or ~3.49e-07	Budget explode or no gain
Apparent "win" then reconfirm (ob20)	3.194e-07 → **3.324e-07**	**Noise**; do not swap
Mild-open / RATE_MIN sweeps (ob21–23)	1.70e-06 … 3.52e-07	Still residual asymptote

Kill: under the official one-core λ, residual throughput cannot stably beat flopscope-native Hadamard. Public ~e-9 scores with `instrumented_share` ≪ `0.001` are a different accounting regime; we do not chase them for the algorithmic write-up.

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4. Relation to cumulant / moment-chain work

Public write-up #314695 shows deep moment chains are *error-compensating dynamical systems*: trajectory-fitted (DAgger) corrections help, step-exact Edgeworth can hurt. We ported the Hermite *K*=10 portable step +

stabilizer into `experiments/moment_chain/`, reproduced the qualitative gap on Mini, and closed two reopen axes locally:

1. **Two-pass residual / absolute refit** on the frozen pass-1 rollout is redundant (pass-2 ≡ pass-1) with both Mini MC-covariance teachers and atlas true M_{11}.
2. **True-cumulant ceiling** (atlas connected pre-activation cumulants replacing the N=4096 probe) improves baked-coef raw MSE by ~1.5× to **~2.9e-06**, matching the public "probe noise costs ~1.5×" claim — and still leaves a **~3e-06 family floor** under linear per-layer heads, far above budgeted Hadamard.

We therefore treat moment chains as a **Phase-2 mechanistic axis**, not a Phase-1 leaderboard contender under current closures.

Our contribution relative to that line is complementary: a **sampling-side error budget** showing that, once variance is driven down with an ONB antipodal design, there is **no cheap residual left for white-box correctors at L31**.

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5. What we are *not* claiming

- We do not claim Hadamard sampling is optimal among all estimators — only that within *budgeted Monte Carlo* at depth 32, ONB+antipodes+χ-radius saturates the gains available from direction design and aggregation.
- We do not claim grey-area residual-wall bypasses (raw ndarray / bundled BLAS) are desirable; after flopscope v0.10's one-core residual safeguard (forum #18125), that channel is intentionally closed (Section 3.5). Our shipped estimator `#325459` is flopscope-native.
- Mini-adjusted improvements of ≤2% without online confirmation are treated as noise (multiple documented Mini↑/online↓ failures; also one residual false win that failed reconfirm).

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6. Reproducibility

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# Environment (Phase-1 evaluator)

pip install -U "flopscope>=0.10.0" "whestbench>=0.14.0"



# Official Mini gate

whest run --estimator estimator.py \

  --dataset hf://aicrowd/arc-whestbench-public-2026@v1-phase1 \

  --split mini --runner subprocess
```

```
# Geometry / aggregation probes (examples)

python experiments/probes/probe_hada_sign_focused.py

python experiments/probes/probe_hada_partial_onb.py

python experiments/moment_chain/eval_mini_gate.py
```

Artifacts: `experiments/probes/results/*_final.json`, `experiments/moment_chain/artifacts/`, development log `DEVELOPMENT.md`.

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7. Phase-1 nominations

Priority	Submission	Role
1	**#325459**	Canonical `pure-0.2.1s` on current evaluator
2	**#325112**	Same estimator; earlier graded confirmation

Write-up deadline for Algorithmic Contribution: **17 August 2026, 23:59 UTC** (extended in forum #18125; earlier #18041 text said Aug 7 — use the extension).

Posting drafts: `writeups/POSTING_DRAFTS.md` (forum body + email to `arc-whestbench@aicrowd.com`).

References (challenge-internal)

1. Submission #325459 / #325112 / #324164 — `pure-0.2.1s` Hadamard+radius.
2. Forum #18097 — trajectory-calibrated moment chain (#314695).
3. Forum #18085 — variance/bias budget for deep white-box means (#317660).
4. Forum #18125 — flopscope v0.10, one-core residual safeguard.
5. Forum #18041 — algorithmic contribution judging guidelines.